CM5 socket, M.2 connector

File: cm5_socket.kicad_sch

System MCU, Watchdog, EEPROM

File: system_mcu.kicad_sch

AC power, DC power, AC sync

File: ac_dc_power.kicad_sch

Isolated digital IO, Fan controller, Encoder

File: digital_io_fan_encoder.kicad_sch

7-port USB3 Hub

File: usb3_hub.kicad_sch

Rear USB, USB-C

File: usb_conn.kicad_sch

Ethernet, Video outputs, Ref clk

File: video_outputs.kicad_sch

DIB 2.0

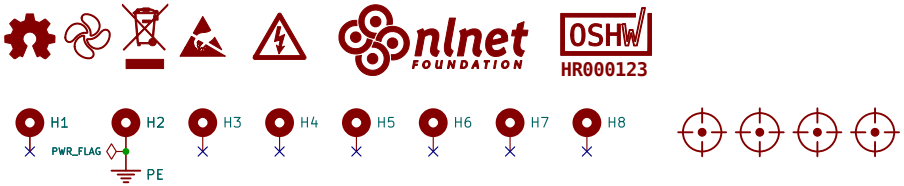
File: dib.kicad_sch

Trigger switch matrix, Power modules coupling

File: power_coupling.kicad_sch

USB Port	USB type	STM32H7 CM5	Raspberry PI CM5
Rear top	USB A	FS only	USB #0, HS/SS
Rear bottom	USB A	FS/HS	USB Hub P1 Flex
Front	USB C	USB Hub P2, HS/SS, Dual role	
DIB #1	DIB 2.0	USB Hub P7, HS/SS	
DIB #2	DIB 2.0	USB Hub P6, HS/SS	
DIB #3	DIB 2.0	USB Hub P5, HS/SS	
DIB #4	DIB 2.0	USB Hub P4, HS/SS	
DIB #5	DIB 2.0	USB Hub P3, HS/SS	

I2C address	Device
0x53	EEPROM M24C32
0x2F	Fan controller EMC2301
0x2D	USB3 Hub
0x3A	PTN5150A USB CC controller
0x70	ADG2188Y 8 x 8 switch array

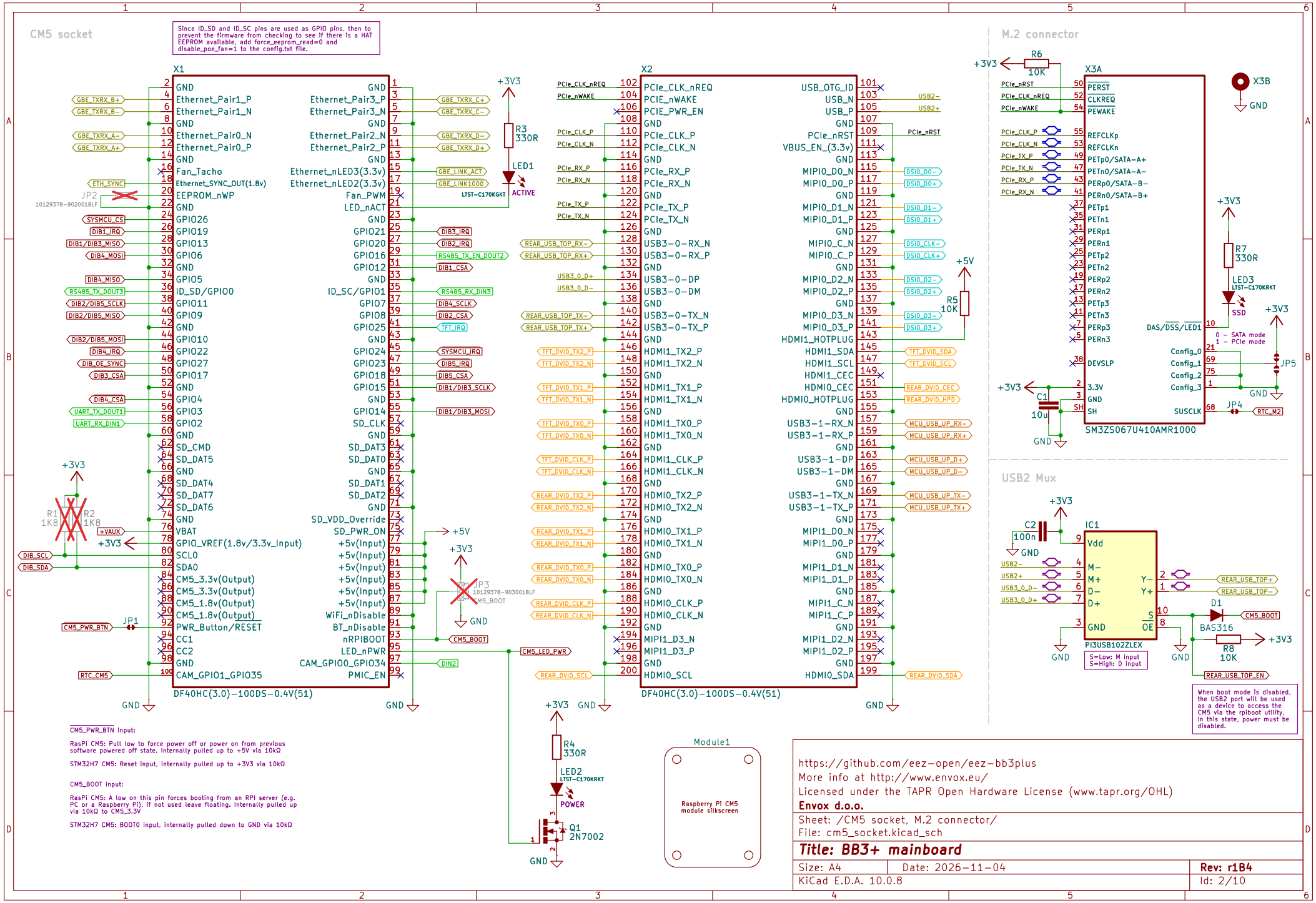


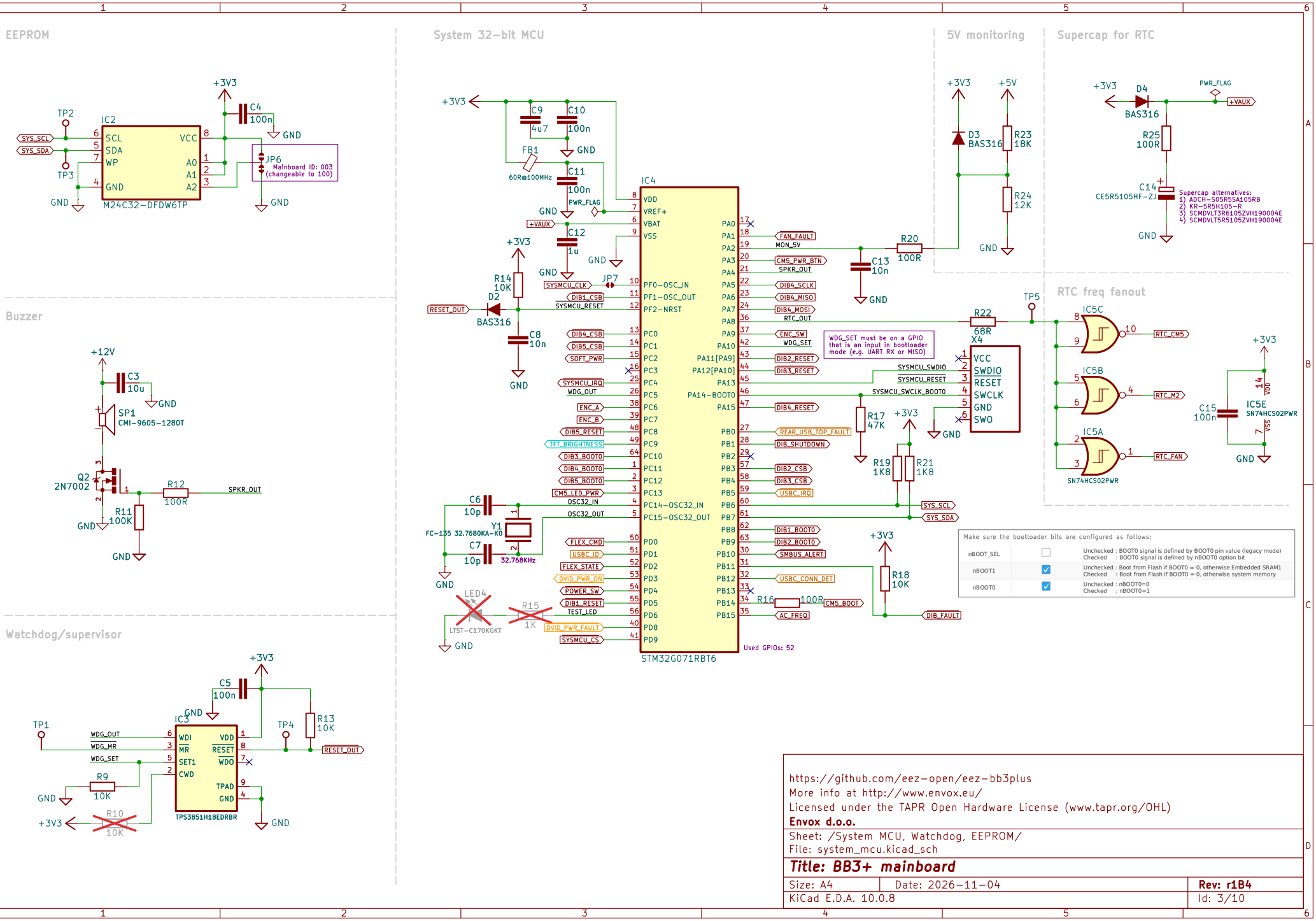
https://github.com/eez-open/eez-bb3plus
More info at http://www.envox.eu/
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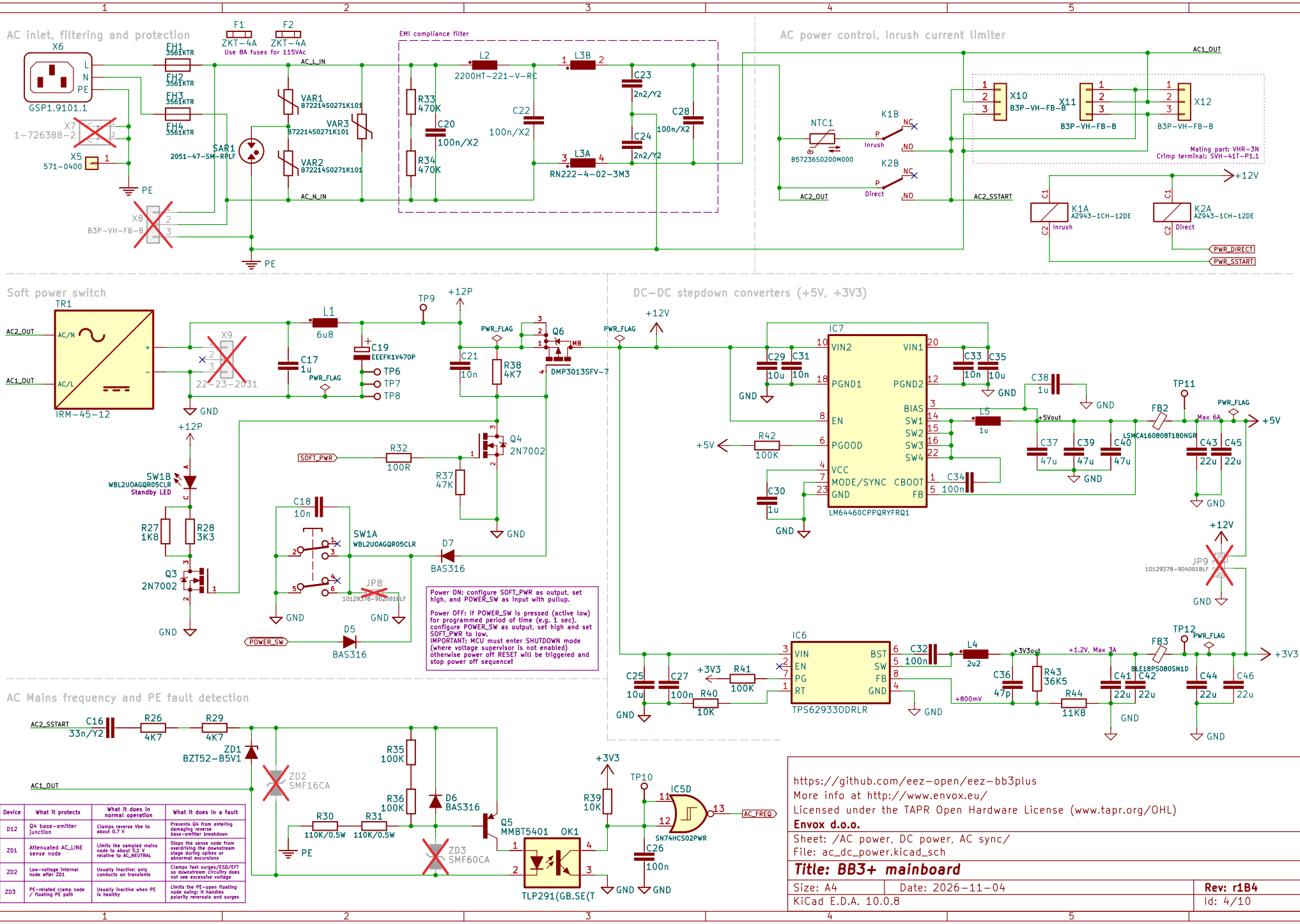
Sheet: /
File: BB3plus mainboard.kicad_sch

Title: BB3+ mainboard

Size: A4	Date: 2026-11-04	**Rev: r1B4**
KiCad E.D.A. 10.0.8		Id: 1/10


CM5 socket
Since ID_SD and ID_SC pins are used as GPIO pins, then to prevent the firmware from checking to see if there is a HAT EEPROM available, add force_eeprom_read=0 and disable_poe_fan=1 to the config.txt file.
X1
DF40HC(3.0)-100DS-0.4V(51)
X2
DF40HC(3.0)-100DS-0.4V(51)
M.2 connector
X3A
SM3ZS067U410AMR1000
X3B
USB2 Mux
IC1
PI3USB102ZLEX
S=Low: M input
S=High: D input
When boot mode is disabled, the USB2 port will be used as a device to access the CM5 via the rpiboot utility. In this state, power must be disabled.
CM5_PWR_BTN input:
RasPi CM5: Pull low to force power off or power on from previous software powered off state. Internally pulled up to +5V via 10kΩ
STM32H7 CM5: Reset input, internally pulled up to +3V3 via 10kΩ
CM5_BOOT input:
RasPi CM5: A low on this pin forces booting from an RPI server (e.g. PC or a Raspberry Pi). If not used leave floating. Internally pulled up via 10kΩ to CM5_3.3V
STM32H7 CM5: BOOT0 input, internally pulled down to GND via 10kΩ
Module1
Raspberry Pi CM5 module silkscreen
https://github.com/eez-open/eez-bb3plus
More info at http://www.envox.eu/
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Sheet: /CM5 socket, M.2 connector/
File: cm5_socket.kicad_sch
Title: BB3+ mainboard
Size: A4
Date: 2026-11-04
Rev: r1B4
KiCad E.D.A. 10.0.8
Id: 2/10

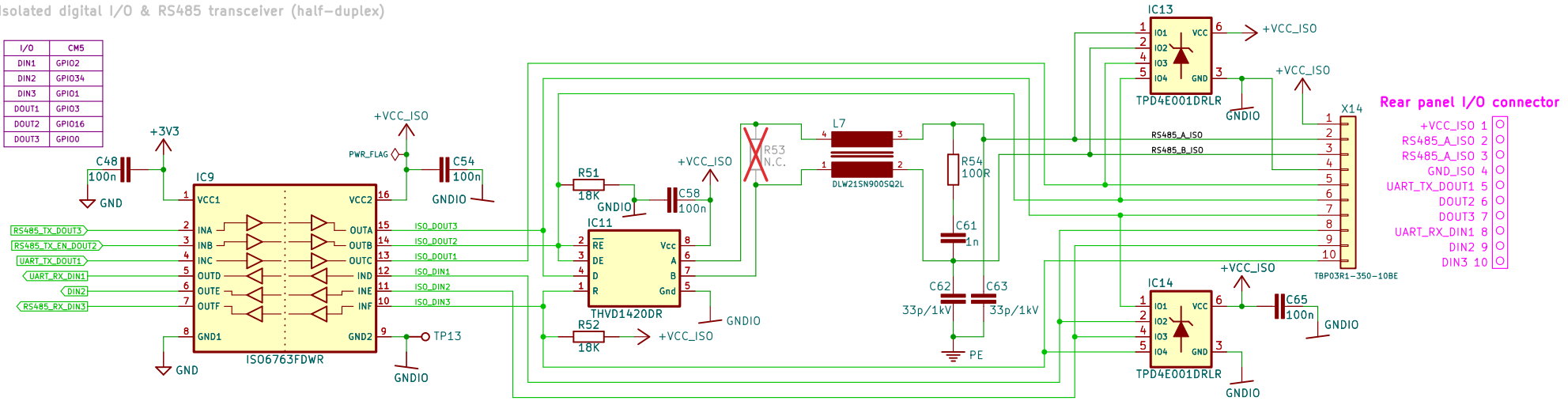

EEPROM
IC2
M24C32-DFDW6TP
JP6
Mainboard ID: 003
(changeable to 100)
Buzzer
SP1
CMI-9605-1280T
Q2
2N7002
Watchdog/supervisor
IC3
TPS3851H18EDRBR
System 32-bit MCU
IC4
STM32G071RBT6
Used GPIOs: 52
WDG_SET must be on a GPIO that is an input in bootloader mode (e.g. UART RX or MISO)
Make sure the bootloader bits are configured as follows:
nBOOT_SEL
nBOOT1
nBOOT0
Unchecked : BOOT0 signal is defined by BOOT0 pin value (legacy mode)
Checked : BOOT0 signal is defined by nBOOT0 option bit
Unchecked : Boot from Flash if BOOT0 = 0, otherwise Embedded SRAM1
Checked : Boot from Flash if BOOT0 = 0, otherwise system memory
Unchecked : nBOOT0=0
Checked : nBOOT0=1
5V monitoring
Supercap for RTC
Supercap alternatives:
1) ADCH-S05R5SA105RB
2) KR-5R5H105-R
3) SCMDVLT3R6105ZVH190004E
4) SCMDVLT5R5105ZVH190004E
RTC freq fanout
SN74HCS02PWR
https://github.com/eez-open/eez-bb3plus
More info at http://www.envox.eu/
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Sheet: /System MCU, Watchdog, EEPROM/
File: system_mcu.kicad_sch
Title: BB3+ mainboard
Size: A4
Date: 2026-11-04
Rev: r1B4
KiCad E.D.A. 10.0.8
Id: 3/10



Device	What it protects	What it does in normal operation	What it does in a fault
D12	Q4 base–emitter junction	Clamps reverse Vbe to about 0.7 V	Prevents Q4 from entering damaging reverse base–emitter breakdown
ZD1	Attenuated AC_LINE sense node	Limits the sampled mains node to about 5.1 V relative to AC_NEUTRAL	Stops the sense node from overdriving the downstream stage during spikes or abnormal excursions
ZD2	Low-voltage internal node after ZD1	Usually inactive; only conducts on transients	Clamps fast surges/ESD/EFT so downstream circuitry does not see excessive voltage
ZD3	PE-related clamp node / floating PE path	Usually inactive when PE is healthy	Limits the PE–open floating node swing; it handles polarity reversals and surges

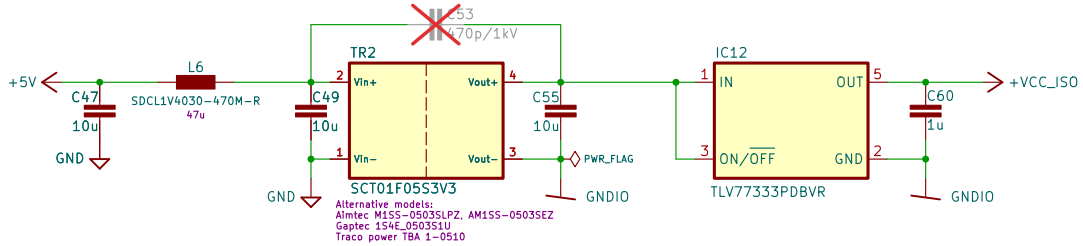
Isolated digital I/O & RS485 transceiver (half-duplex)

I/O	CM5
DIN1	GPIO2
DIN2	GPIO34
DIN3	GPIO1
DOUT1	GPIO3
DOUT2	GPIO16
DOUT3	GPIO0

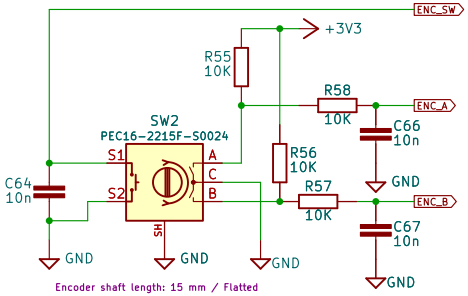


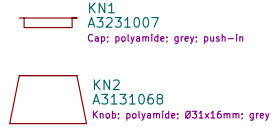
Rear panel I/O connector

Isolated regulated power supply (+5V)

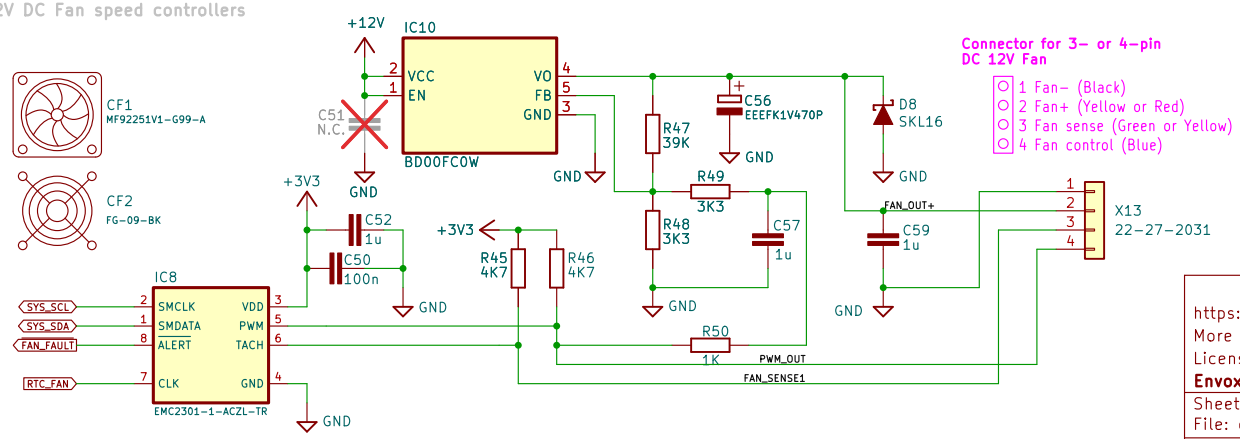


Encoder with switch





12V DC Fan speed controllers



https://github.com/eez-open/eez-bb3plus
More info at http://www.envox.eu/
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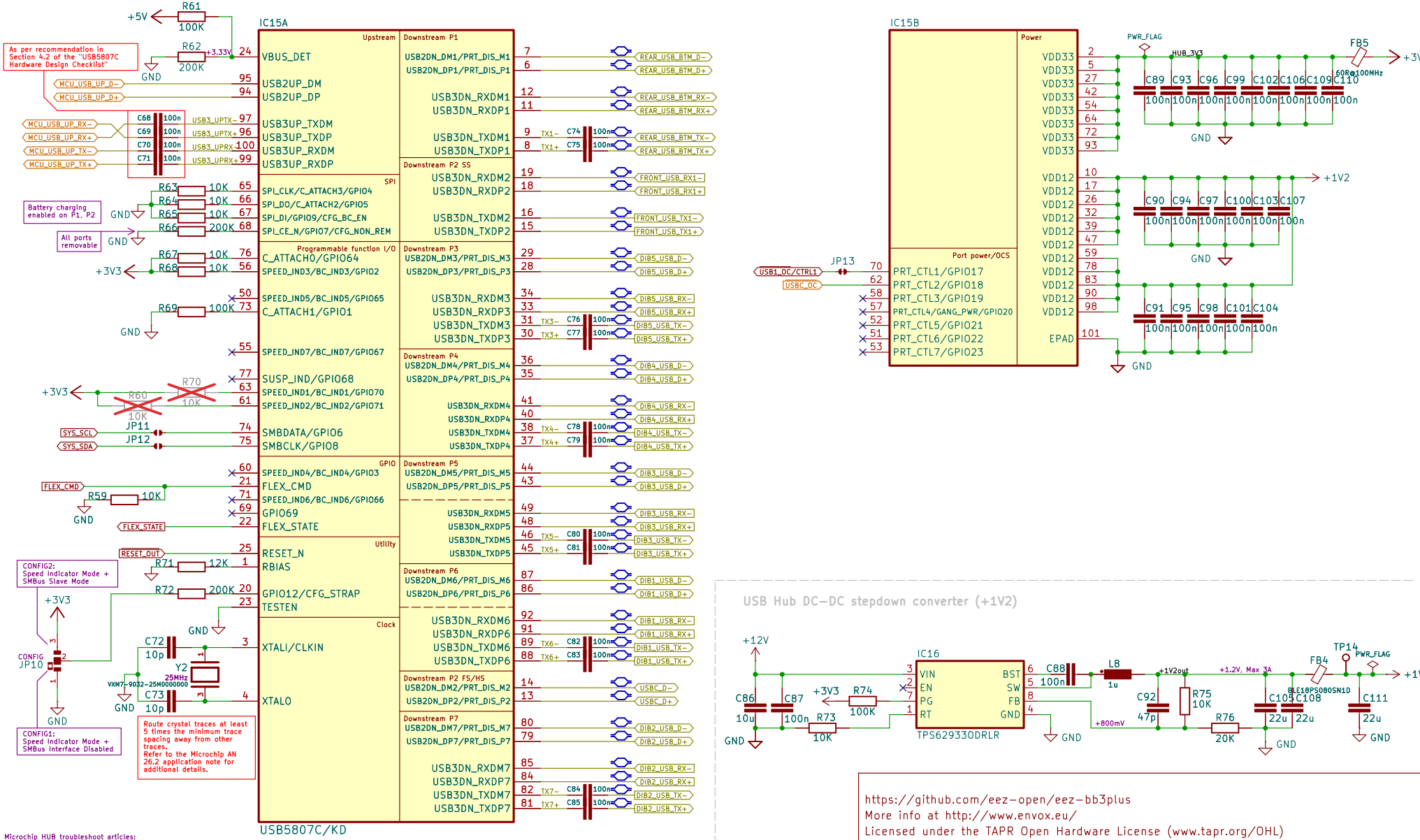
Sheet: /Isolated digital IO, Fan controller, Encoder/
File: digital_io_fan_encoder.kicad_sch

Title: BB3+ mainboard

Size: A4	Date: 2026-11-04	**Rev: r1B4**
KiCad E.D.A. 10.0.8		Id: 5/10

7-port USB hub

All downstream ports that are not used should be disabled (use I2C to configure).



Microchip HUB troubleshoot articles:

https://support.microchip.com/s/article/USB3-Hub-Upstream-Port-Connect-Enumeration-Failure
https://support.microchip.com/s/article/USB-High-Speed-Chirp-Explained
https://support.microchip.com/s/article/USB3-Connection-Process-and-Attach-Detection

IMPORTANT:

If I2C/SMBus pull-up resistors are detected by the USB5807C at start-up, the hub will wait indefinitely to be configured by the attached I2C/SMBus master. For early prototyping, it may be necessary to physically remove the pull-up resistors until the I2C/SMBus master is fully operational and able to properly configure the hub at start-up.

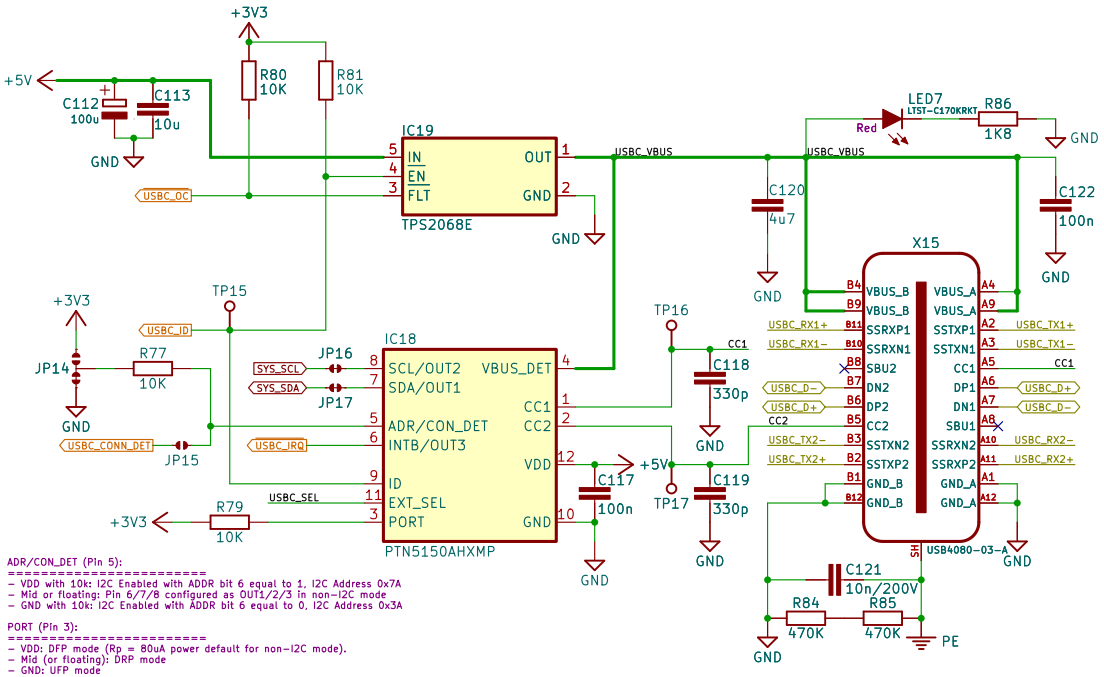
USB Hub DC-DC stepdown converter (+1V2)

https://github.com/eez-open/eez-bb3plus
More info at http://www.envox.eu/
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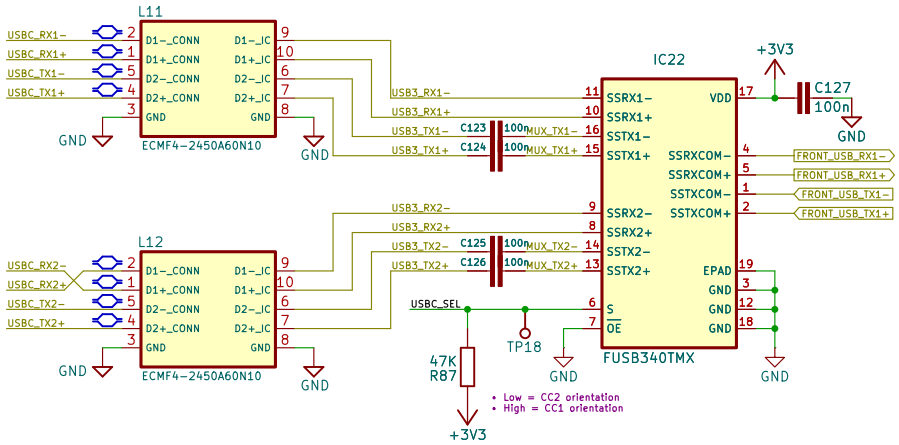
Envox d.o.o.

Sheet: /7-port USB3 Hub/		
File: usb3_hub.kicad_sch		
Title: BB3+ mainboard		
Size: A4	Date: 2026-11-04	**Rev: r1B4**
KiCad E.D.A. 10.0.8		Id: 6/10

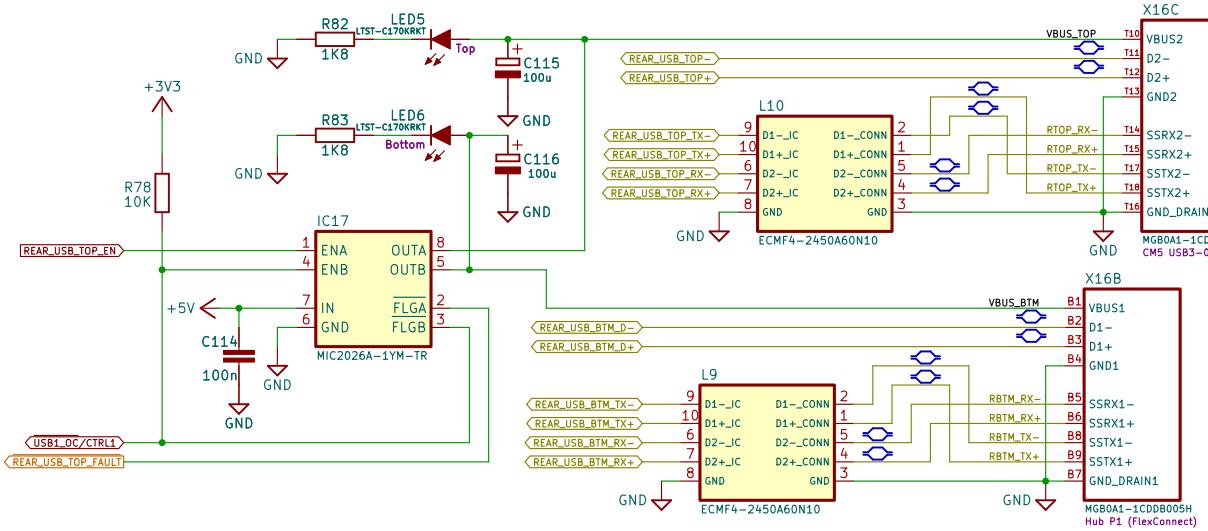
USB-C 3.0 front port (source max 1.5A)



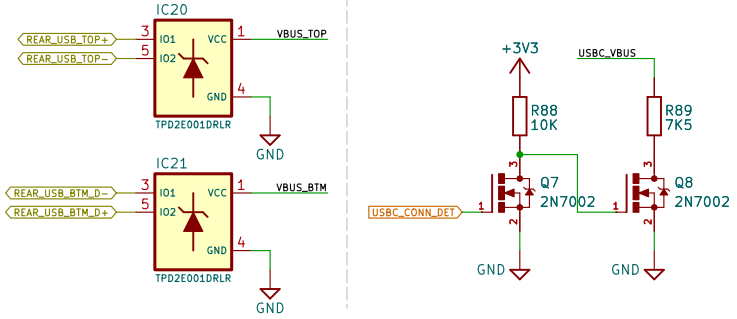
USB-C 3.0 Protection and Mux



USB2.0 Type-A (Rear panel)



VBus Discharge Circuit



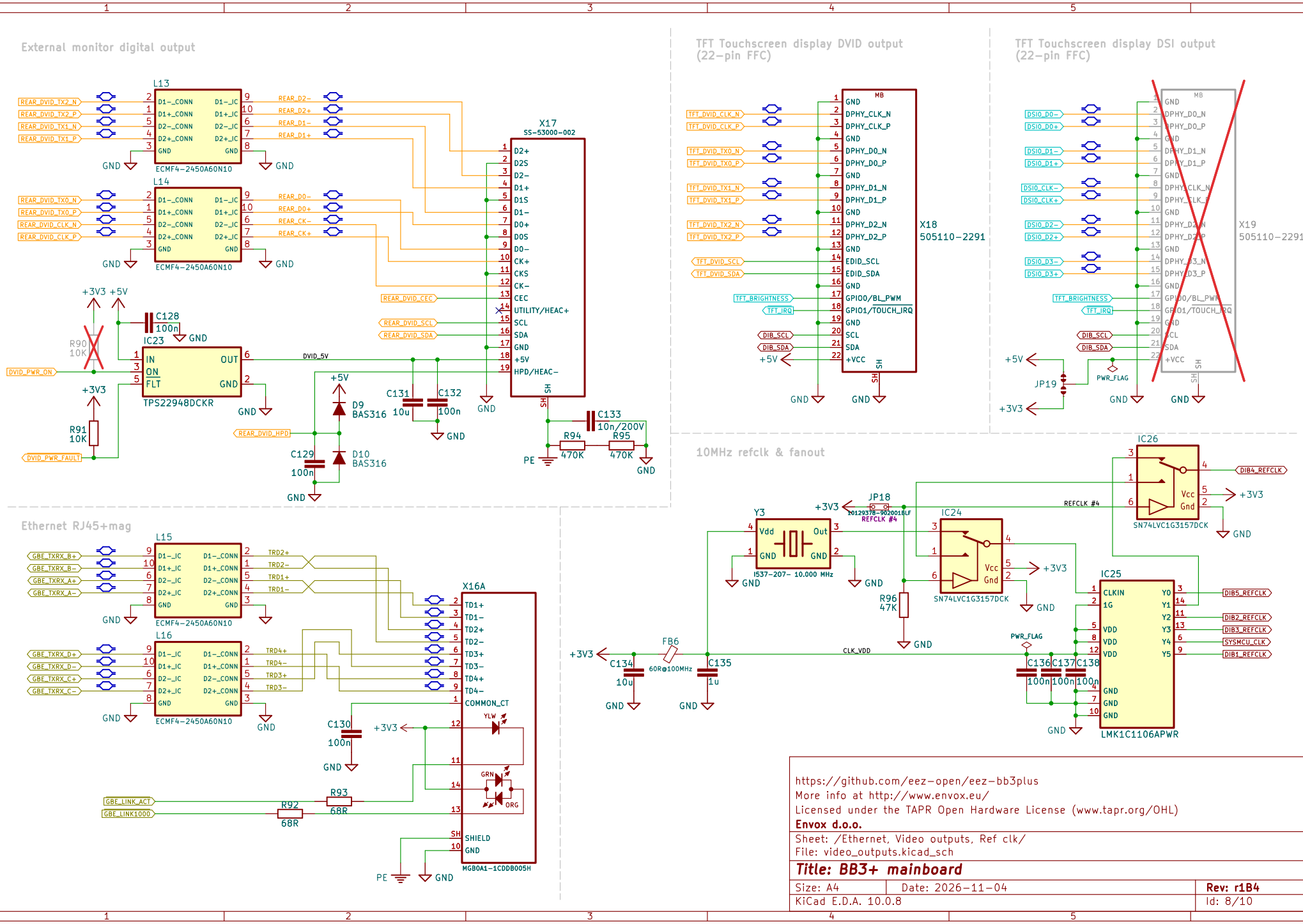
https://github.com/eez-open/eez-bb3plus
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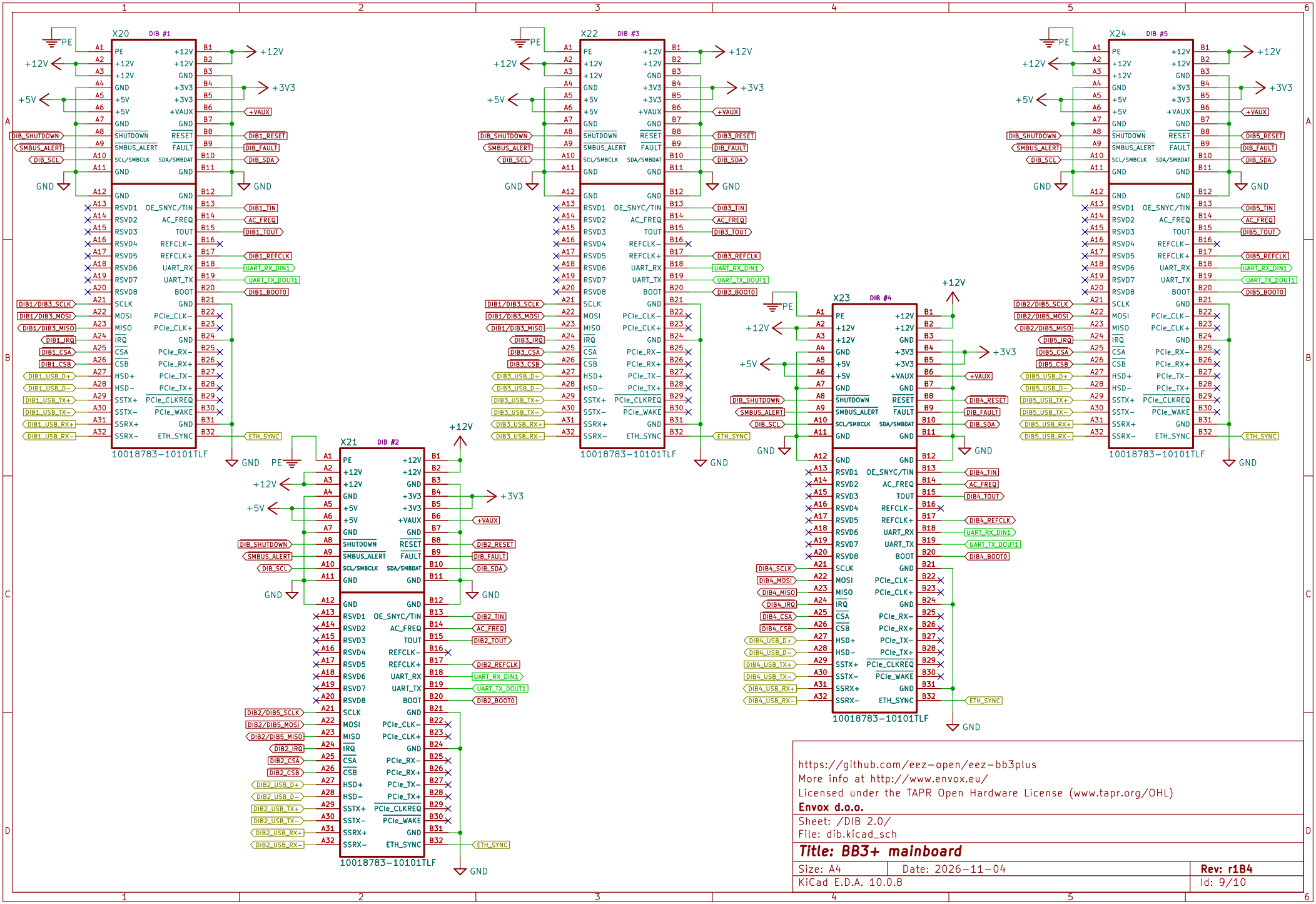
Envox d.o.o.

Sheet: /Rear USB, USB-C/
File: usb_conn.kicad_sch

Title: BB3+ mainboard

Size: A4	Date: 2026-11-04	**Rev: r1B4**
KiCad E.D.A. 10.0.8		Id: 7/10


External monitor digital output
L13
ECMF4-2450A60N10
L14
ECMF4-2450A60N10
X17
SS-53000-002
IC23
TPS22948DCKR
C128
100n
R90
10K
R91
10K
C131
10u
C132
100n
C129
100n
D9
BAS316
D10
BAS316
C133
10n/200V
R94
470K
R95
470K
TFT Touchscreen display DVID output
(22-pin FFC)
X18
505110-2291
TFT Touchscreen display DSI output
(22-pin FFC)
X19
505110-2291
JP19
PWR_FLAG
Ethernet RJ45+mag
L15
ECMF4-2450A60N10
L16
ECMF4-2450A60N10
X16A
MGB0A1-1CDDB005H
C130
100n
R92
68R
R93
68R
10MHz refclk & fanout
IC26
SN74LVC1G3157DCK
JP18
Y3
I537-207- 10.000 MHz
IC24
SN74LVC1G3157DCK
R96
47K
FB6
60R@100MHz
C134
10u
C135
1u
IC25
LMK1C1106APWR
C136
100n
C137
100n
C138
100n
https://github.com/eez-open/eez-bb3plus
More info at http://www.envox.eu/
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Envox d.o.o.
Sheet: /Ethernet, Video outputs, Ref clk/
File: video_outputs.kicad_sch
Title: BB3+ mainboard
Size: A4
Date: 2026-11-04
Rev: r1B4
KiCad E.D.A. 10.0.8
Id: 8/10



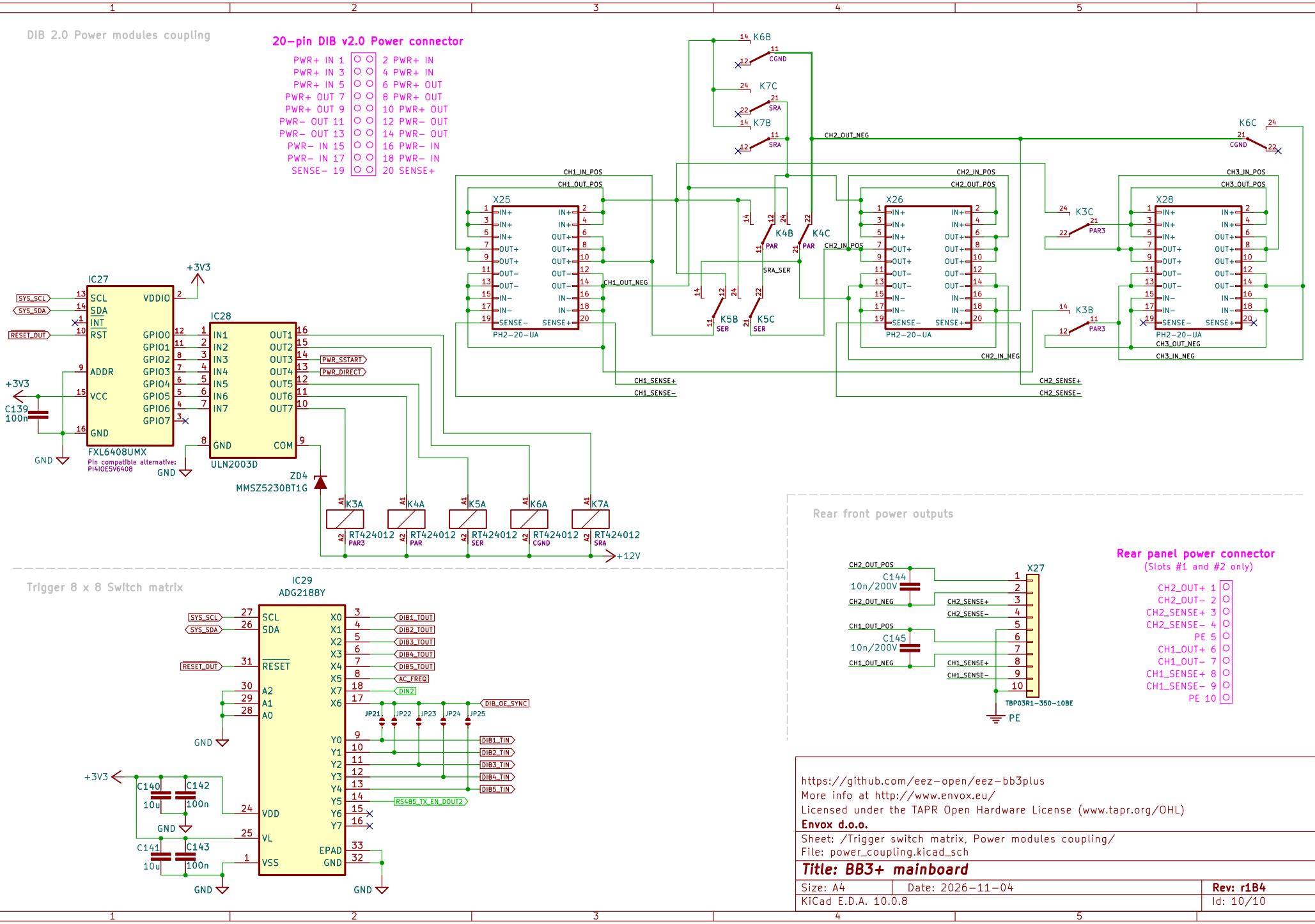
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Sheet: /DIB 2.0/
File: dib.kicad_sch

Title: BB3+ mainboard

Size: A4	Date: 2026-11-04	**Rev: r1B4**
KiCad E.D.A. 10.0.8		Id: 9/10


DIB 2.0 Power modules coupling
20-pin DIB v2.0 Power connector
PWR+ IN 1
2 PWR+ IN
PWR+ IN 3
4 PWR+ IN
PWR+ IN 5
6 PWR+ OUT
PWR+ OUT 7
8 PWR+ OUT
PWR+ OUT 9
10 PWR+ OUT
PWR- OUT 11
12 PWR- OUT
PWR- OUT 13
14 PWR- OUT
PWR- IN 15
16 PWR- IN
PWR- IN 17
18 PWR- IN
SENSE- 19
20 SENSE+
K6B
K7C
K7B
K6C
CGND
SRA
CH2_OUT_NEG
CH1_IN_POS
CH1_OUT_POS
CH2_IN_POS
CH2_OUT_POS
CH3_IN_POS
CH3_OUT_POS
X25
X26
X28
PH2-20-UA
K4B
K4C
K3C
K3B
PAR
PAR3
SRA_SER
K5B
K5C
SER
CH1_OUT_NEG
CH2_IN_NEG
CH3_OUT_NEG
CH3_IN_NEG
CH1_SENSE+
CH1_SENSE-
CH2_SENSE+
CH2_SENSE-
IC27
FXL6408UMX
Pin compatible alternative:
PI4IOE5V6408
IC28
ULN2003D
+3V3
SYS_SCL
SYS_SDA
RESET_OUT
C139
100n
GND
PWR_SSTART
PWR_DIRECT
ZD4
MMSZ5230BT1G
K3A
K4A
K5A
K6A
K7A
RT424012
PAR3
PAR
SER
CGND
SRA
+12V
Trigger 8 x 8 Switch matrix
IC29
ADG2188Y
DIB1_TOUT
DIB2_TOUT
DIB3_TOUT
DIB4_TOUT
DIB5_TOUT
AC_FREQ
DIN2
DIB_OE_SYNC
JP21
JP22
JP23
JP24
JP25
DIB1_TIN
DIB2_TIN
DIB3_TIN
DIB4_TIN
DIB5_TIN
RS485_TX_EN_DOUT2
C140
C142
C141
C143
10u
100n
Rear front power outputs
C144
C145
10n/200V
X27
TBP03R1-350-10BE
PE
Rear panel power connector
(Slots #1 and #2 only)
CH2_OUT+ 1
CH2_OUT- 2
CH2_SENSE+ 3
CH2_SENSE- 4
PE 5
CH1_OUT+ 6
CH1_OUT- 7
CH1_SENSE+ 8
CH1_SENSE- 9
PE 10
https://github.com/eez-open/eez-bb3plus
More info at http://www.envox.eu/
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Envox d.o.o.
Sheet: /Trigger switch matrix, Power modules coupling/
File: power_coupling.kicad_sch
Title: BB3+ mainboard
Size: A4
Date: 2026-11-04
Rev: r1B4
KiCad E.D.A. 10.0.8
Id: 10/10